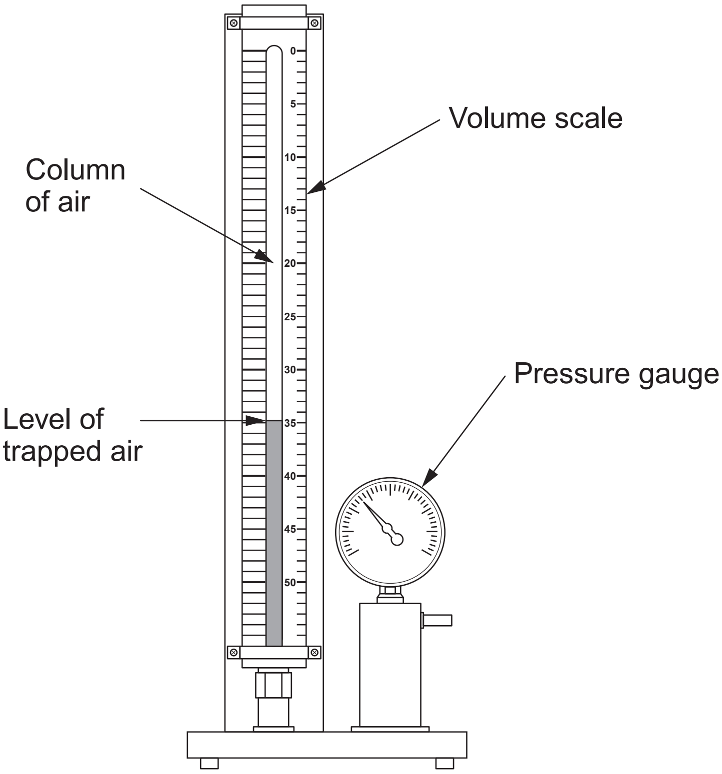
Examiner only

8. (a) The volume of a gas is related to its pressure.
A mass of air is contained in a glass tube that is attached to a pressure gauge.



(i) When the pressure shown by the pressure gauge increases and the temperature is constant, the volume of the column of air decreases.
Describe the effect, if any, on the motion of the molecules and on the collisions with the container. [2]

....

....

....

(ii) At a pressure of 1.2×10^5 Pa, the volume is 18 cm^3.
Explain, without calculation, what happens to the volume when the pressure is increased to 2.4×10^5 Pa at constant temperature. [2]

....

....

....

Examiner only

(b) In another experiment, air initially at 27 °C is kept at constant volume.
The air pressure is increased from 2.4×10^5 Pa, to 3.0×10^5 Pa.

(i) State the initial temperature in Kelvin. [1]

Temperature = ... K

(ii) Use an equation from page 2 to calculate the **change in temperature** of the air. [3]

Change in temperature = ... K

(c) A mass of 5×10^{-3} kg of air requires 105.5 J of thermal energy to increase its temperature by 21 °C.

Use the equation:

$$\Delta Q = mc\Delta\theta$$

to calculate the specific heat capacity, c, of air. [3]

Specific heat capacity = ... J / kg °C

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END OF PAPER